02 / ALGEBRA / CLASS_17_VECTORS / EX26

EX26 exercise guide

Algebra practice, manual verification and 2D mechanical work.

DOCUMENT 03

Learning objectives

- Complete an algebra section and a physics section.
- Design and implement the steps yourself.
- Check results against expected values and units.

Prerequisites: Prerequisites: EX25 completed, NumPy arrays, vector components and Euclidean norm.

Reading order: 01 Theory → 02 Reference → 03 Exercise → 04 Pseudocode → Coding.

1. Mission and constraints

EX26 develops the dot product in two stages: **Part A, algebra** (no units), and **Part B, mechanical work** (N, m and J). You will implement the operations yourself; check values do not replace calculations.

> Use a single .py script, NumPy, variables, expressions and terminal output. No JSON, pytest or src structure. Simple functions are optional, not required.

2. Part A — unitless algebra

A1. Worked manual example (for studying)

Consider the vectors:

$$\mathbf{u} = (2, 1), \qquad \mathbf{v} = (1, 3)$$
$$\mathbf{u} \cdot \mathbf{v} = 2(1) + 1(3) = 5$$
$$\|\mathbf{u}\|_2 = \sqrt{5}, \quad \|\mathbf{v}\|_2 = \sqrt{10}$$
$$\theta = \arccos\left(\frac{5}{\sqrt{50}}\right) = 45\,^\circ$$

This example is already solved manually. Use it to understand the steps and verify the theory.

A2. Your algebra task (code it)

Given vectors **a** and **b**, implement the following without copying a complete solution:

$$\mathbf{a} = (3, -1), \qquad \mathbf{b} = (2, 2)$$

- Create NumPy arrays for a and b.
- Compute a·b by components on paper and then with np.dot().
- Calculate both Euclidean norms.
- Find cos θ and the angle between the vectors in degrees.
- Verify by substituting values into the geometric formula.

A3. Orthogonality: a second check

$$\mathbf{p} = (2, 1), \qquad \mathbf{q} = (1, -2)$$

Calculate p·q and determine whether the vectors are perpendicular. Explain why you obtain 90° and why the dot product is not a vector.

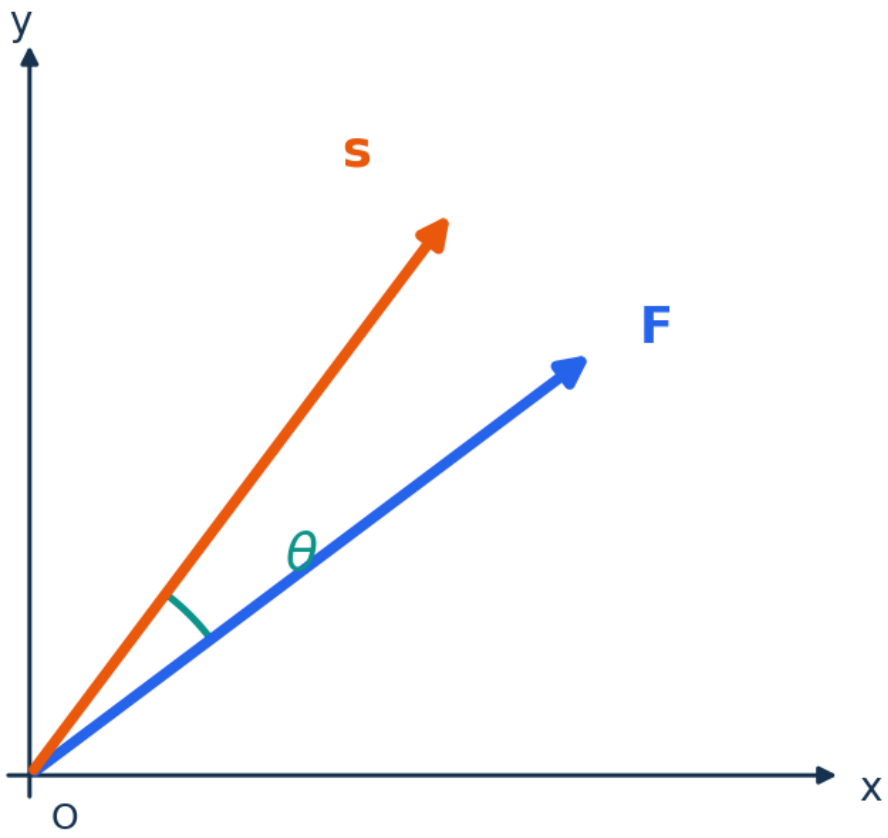
3. Part B — engineering: mechanical work

A constant force acts on a body moving in a plane. Express both vectors by components relative to the Cartesian unit vectors.

$$\hat{\mathbf{i}} = (1, 0), \qquad \hat{\mathbf{j}} = (0, 1)$$
$$\mathbf{F} = (8\hat{\mathbf{i}} + 6\hat{\mathbf{j}})\ \text{N}$$

$$\mathbf{s} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}})\ \text{m}$$

Equivalently, **F** = (8, 6) N and **s** = (3, 4) m. The coefficients are components; hatted i and j are unit vectors, not the imaginary unit.



Force and displacement directions (schematic; N and m have no common scale).

- Create arrays for force and displacement.
- Compute mechanical work with the dot product.
- Calculate force and displacement magnitudes.
- Determine cosine and angle in degrees.
- Interpret the work sign and use correct units.

B2. Two physical cases for consolidation

Without changing the general calculation, repeat it for these vector pairs and interpret the sign of work:

$$\mathbf{F}_{\perp} = (10, 0)\ \text{N}, \quad \mathbf{s}_{\perp} = (0, 2)\ \text{m}$$
$$\mathbf{F}_{\text{opp}} = (-5, 0)\ \text{N}, \quad \mathbf{s}_{\text{opp}} = (4, 0)\ \text{m}$$

4. Check values and validation

Use these as references for verification, not variables to insert in place of computed results.

a · b	4
$\|\|a\|\|_2$	3.1623
$\|\|b\|\|_2$	2.8284
cos θ (A2)	0.4472
θ (A2)	63.4349°
p · q	0
θ (A3)	90°
W (B1)	48 J
θ (B1)	16.2602°
W (B2)	0 J
W (B3)	−20 J

5. Interpretation questions

- Why does the dot product return a number rather than a vector?
- What is the physical meaning of W > 0, W = 0 and W < 0?
- What fails when calculating the angle of a zero vector?
- Why does force in N multiplied by displacement in m give J?

6. File location and execution

EX26 belongs to the **same class 17** as EX25: do not create another class folder. Add a second script.

```
Python_Engineering/
└── 02_numerical_methods/
    └── algebra/
        └── class_17_vectors/
            ├── EX25_vectors_and_forces.py
            ├── EX26_dot_product.py      <-- NUEVO / NEW
            └── docs/
                ├── ES/
                └── EN/
```

In the VS Code terminal, while inside class_17_vectors, run:

```
python EX26_dot_product.py
```

7. Programming hints (at the end, without full solution)

The following expressions are isolated syntax examples, not a complete solution. Design the algorithm yourself.

```
import numpy as np

x = np.array([1.0, 2.0])
y = np.array([0.0, 3.0])

print(np.dot(x, y))           # 6.0
print(np.linalg.norm(x))      # sqrt(5)
```

For an angle you need norms, the dot product, np.arccos and np.degrees. The Reference Guide explains their inputs and outputs.